

Sl. No. of Ques. Paper : 5719 F
Unique Paper Code : 234403
Name of Paper : (CSHT-409) Data Communication and Computer Networks
Name of Course : B.Sc. (Hons.) Computer Science
Semester : IV
Duration : 3 hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

The paper has two Sections. All questions in Section A are compulsory. Attempt any four questions from Section B. Subparts of the questions should be answered together.

Section-A
(All Questions are compulsory)

- Q. 1.(a) Draw the pulse diagram for bit stream 1110010100111, for the following encoding techniques
(i) RZ (ii) Manchester (iii) Differential Manchester [6]
- (b) Why are communication satellites in geosynchronous orbits? What is the prime advantage of LEOS. [4]
- (c) What do these special IP address denote: [3]
(i) 0.0.0.0
(ii) 255.255.255.255
(iii) 127.xx.yy.zz
- (d) Write the port number of following protocols: [3]
FTP, TELNET, HTTP
- (e) Differentiate between Adaptive and Non- Adaptive algorithms. Give Example of each? [4]
- (f) What is window size, how pipelining property is used in sliding window protocols? [4]
- (g) What is meant by Checksum and how is it useful in networks? [3]
- (h). Give four characteristics of point to point protocol. [4]
- (i) Find the minimum bandwidth for an FSK signal transmitting at 2000bps. Transmission is in half-duplex mode and the carriers are separated by 3000HZ. [2]
- (j) What are two features of HTTP protocol ? [2]

Section -B.
(Attempt any four questions from section B)

- Q.2. (a) Explain TCP protocol Header with the help of a diagram. [6]
- (b) What are the minimum and maximum frame sizes for Ethernet frames? Why can't the minimum frame length be zero. [4]

Turn over

- Q.3. (a) Discuss the collision detection process in a CSMA/CD protocol can a station transmit and receive frames at the same time. [5]
- (b) Briefly explain what is the important role of RPC in the communication network. [5]
- Q.4. (a) What are the operational difference between ARP and RARP. [4]
- (b) A router inside an organization receives a packet with the destination address 190.240.34.95. If the subnet mask is /19, find the subnet address. [2]
- (c) How do multiple senders share the common transmission media in a network? List three techniques commonly used for this purpose. [4]
- Q.5 Explain the differences between circuit switching and packet switching. How is virtual circuit approach different from datagram approach, briefly explain? [10]
- Q.6. (a) Explain following terms: [2*5]
- (i) Cladding
 - (ii) Flow Control
 - (iii) Flooding
 - (iv) URL
 - (v) BOOTP
- Q.7. (a) The following character encoding is used in a data link protocol: [6]
 A: 01000111; B: 11100011; FLAG: 01111110; ESC: 11100000 show the bit sequence Transmitted (in binary) for the four character frame: A B ESC FLAG when each of the following framing methods are used:
- (i) Character count
 - (ii) Flag bytes with byte stuffing.
 - (iii) Starting and ending flag bytes, with bit stuffing.
- (b) Write any two functionalities for each of the following network devices: [2*2]
- (i) Switches
 - (ii) Gateways

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